SUBRATA MALLICK

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PROFESSIONAL SUMMARY

ECE graduate with hands-on experience building full-stack and AI-powered applications using Java, Spring Boot, MySQL, Python, React/Next.js and REST APIs. Experienced in backend architecture, multimodal AI integration, machine learning, database design and production deployment. Seeking an entry-level software development role.

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript, TypeScript, C
Backend: Spring Boot, Spring Data JPA, Hibernate, REST APIs, FastAPI
Frontend: HTML, CSS, JavaScript, React, Next.js, Tailwind CSS, Recharts
Database: MySQL, SQL
AI/ML: scikit-learn, Machine Learning, Face Recognition, InsightFace, Gemini Multimodal API, Prompt Engineering
Tools: Git, GitHub, Gradle, Postman, Swagger/OpenAPI, IntelliJ IDEA
Core: DSA, OOP, DBMS, DTOs, Validation, Exception Handling

EDUCATION

B.Tech — Electronics and Communication Engineering | Academy of Technology | 2022–2026 | CGPA: 7.95
Higher Secondary (12th) | Bhatpara Amarkrishna Pathsala (H.S.) | 2022 | 80.8%
Secondary (10th) | Bhatpara Amarkrishna Pathsala | 2020 | 88.85%

PROJECTS

SchoolBuddy.ai — AI-Powered Multimodal Educational Tutor | Java, Spring Boot, React/Next.js, MySQL, Gemini API

Live: https://schoolbuddy-frontend.vercel.app **Frontend:** https://github.com/MallickSayan407/schoolbuddy-frontend **Backend:** https://github.com/MallickSayan407/schoolbuddy-backend

- Developed a full-stack AI tutoring platform for Grade 1–10 students supporting text and image-based questions with grade-aware responses across Math, Science and English.
- Built a Spring Boot REST backend with MySQL conversation persistence, DTO-based APIs, validation, centralized exception handling and contextual multi-turn conversations.
- Integrated multimodal Gemini AI with a multi-model fallback strategy, grade-specific prompting and learning modes: Explain, Solve, Hint, Simplify, Practice and Quiz.
- Developed a responsive Next.js/React frontend with grade/subject selection, image upload, conversation UI, Markdown/KaTeX rendering and real-time AI interaction.
- Implemented mathematical response normalization and educational prompting for LaTeX formatting, age-appropriate explanations and structured step-by-step solutions.
- Deployed frontend on Vercel, backend on Render and database on Aiven MySQL; production-tested text, mathematical, conversational and image-based workflows.

Cricket Analytics | Java 21, Spring Boot, MySQL, Python, React, JavaScript, Recharts

Live: https://cricket-analytics-two.vercel.app **GitHub:** https://github.com/MallickSayan407/CricketAnalytics

- Built a full-stack cricket analytics platform covering international and IPL data with players, teams, matches, venues, seasons, leaderboards, filtering and visualization.
- Integrated a Python/scikit-learn ODI prediction service through FastAPI and added a Monte Carlo tournament simulator for outcome analysis.

Face Attendance System | Java 21, Spring Boot, MySQL, Python, InsightFace, HTML, CSS, JavaScript

GitHub: https://github.com/MallickSayan407/face-attendance-system

- Developed a full-stack employee attendance system using layered architecture, REST APIs, image upload, attendance records, duplicate prevention and summary reporting.
- Integrated Python InsightFace with Spring Boot and implemented DTOs and centralized exception handling for reliable API workflows.

Employee Management System | Java, Spring Boot, MySQL, HTML, CSS, JavaScript

GitHub: https://github.com/MallickSayan407/employee-management

- Implemented CRUD APIs with DTOs, validation, centralized exception handling, pagination, sorting, search/filtering and Swagger/OpenAPI documentation.

ACADEMIC PROJECT

Substrate Analysis on the Design of Wide-band Antenna for 5G Sub-6 GHz Communication | 2024–2026

TRAINING & CERTIFICATION

Railway Training under Sealdah Division

INTERESTS

Cricket | Football | Recitation